ROLE 4 ORAL DEFENSE & VIVA HANDBOOK • STUDENT RESEARCH TEAM

VIVA & PRESENTATION REFERENCE GUIDE

Complete Technical Defense Preparation: 23 Mandatory Examination Questions, Mathematical Formulations, Architecture Walkthrough, Comprehensive Glossary, and Academic Boundary Safeguards

Project: ML-Based Trust-Aware PoS
Focus: Viva & Examination Defense
Audience: Faculty Examiners & Viva Panel
Key Math: Weight = Stake × Trust
ML Model: Random Forest Classifier
Dataset: 100 Nodes, 1,000 Tx
Date: September 2026
Lead Author: Role 4 (Research Lead)
Format: High-Yield Q&A Guide

Viva Preparation Strategy

This document equips team members to confidently defend the project during oral examinations, project reviews, and viva voce. Every answer is structured to be: (1) *Technically rigorous* (using precise distributed systems and ML terminology), (2) *Concise enough to speak aloud* in 45–60 seconds, and (3) *Defensible against tough examiner cross-examination*.

1. Key Conceptual Formulas Summary

Mechanism	Mathematical Formulation	Operational Interpretation
Traditional PoS Selection	$$P_{\text{trad}}(i) = \frac{\text{Stake}_i}{\sum_{j=1}^N \text{Stake}_j}$$	Proposal probability is strictly proportional to capital collateral. Behaviorally blind.
Trust Score Output	$$T_i = P(\text{Honest} \mid \mathbf{x}_i) = \frac{1}{N_{\text{trees}}} \sum_{t=1}^{N_{\text{trees}}} \mathbb{I}(h_t(\mathbf{x}_i) = 1)$$	Posterior probability of honest protocol fidelity computed across Random Forest ensemble.
Trust-Aware Selection Weight	$$W_i = \text{Stake}_i \times T_i$$	Selection weight couples financial capital multiplicatively with historical operational trust.
Trust-Aware Selection Probability	$$P_{\text{trust}}(i) = \frac{\text{Stake}_i \times T_i}{\sum_{j=1}^N (\text{Stake}_j \times T_j)}$$	Decouples consensus leadership from pure wealth; slashes proposal odds for degraded nodes.
Malicious Selection Rate (MSR)	$$\text{MSR} = \frac{\sum_{k=1}^{N_{\text{slots}}} \mathbb{I}(\text{Proposer}_k \in \text{Malicious})}{N_{\text{slots}}}$$	Primary security benchmark: fraction of consensus proposal slots awarded to rogue nodes.

2. 23 Mandatory Viva Questions & Concise Defensible Answers (Questions 1 to 12)

Q1: What is a blockchain?

Answer: A blockchain is an append-only, decentralized distributed ledger composed of cryptographically linked blocks of transactions. It is maintained across a peer-to-peer network without a central intermediary. Blocks are bound together by

recursive hash pointers, meaning changing data in an earlier block alters its cryptographic digest, invalidating all subsequent blocks and ensuring tamper-evident immutability.

Q2: Why is distributed consensus necessary in blockchain?

Answer: In an open, asynchronous network, nodes do not trust one another, clocks drift, and network partitions occur. Consensus is required to guarantee distributed agreement on a single, deterministic order of state transitions. It prevents double-spending attacks and maintains state machine replication despite crash failures or adversarial Byzantine actions.

Q3: What is the fundamental difference between Proof of Work (PoW) and Proof of Stake (PoS)?

Answer: PoW ties consensus authority to physical energy and hardware expenditure (solving non-invertible SHA-256 puzzles), making Sybil attacks thermodynamically expensive. PoS ties consensus authority to internal financial capital (locking native cryptocurrency tokens as collateral). PoS achieves greater than 99.9% energy reduction and higher throughput, but decouples identity from the physical world.

Q4: How does traditional PoS validator selection work?

Answer: Traditional PoS selects block proposers via a weighted lottery where the probability of selecting validator i is proportional to its relative fraction of total staked collateral ($P(i) = S_i / \sum S_j$). While production systems like Ethereum use discrete effective balances and RANDAO randomness, the conceptual principle remains strictly stake-weighted.

Q5: Why is stake alone insufficient as a signal of validator reliability?

Answer: Stake measures wealth, not operational integrity. A wealthy validator can still suffer hardware misconfigurations, experience severe network latency, censor transactions, or have its private signing keys stolen by an external attacker. Relying exclusively on stake leaves the consensus mechanism blind to active node unreliability and compromised keys.

Q6: What is trust-aware validator selection?

Answer: Trust-aware validator selection is a consensus enhancement where block proposal odds depend on both economic capital and historical operational fidelity. Telemetry from past validation rounds is fed into an algorithm or ML model to generate a trust score ($T_i \in [0, 1]$), which scales the selection weight ($W_i = S_i \times T_i$).

Q7: Why use behavioral features instead of static validator ratings?

Answer: Static credentials or administrative authorizations cannot detect dynamic operational degradation or key compromise. Continuous behavioral telemetry—such as uptime, response latency, verification accuracy, and protocol anomaly flags—provides real-time empirical evidence of how a validator actually executes its duties.

Q8: Why did you choose these specific five features?

Answer: We selected `uptime`, `response_time_ms`, `verification_correct`, `transaction_outcome`, and `suspicious_flag` because they capture the essential pillars of distributed consensus: liveness (uptime and latency), execution safety (verification correctness), censorship resistance (approval/decline outcome), and protocol integrity (anomaly flags). Furthermore, they are grounded in peer-reviewed literature (IEEE ICBC 2026, ACM IECC 2019) and are straightforward to simulate deterministically.

Q9: Why use a Random Forest classifier rather than a Deep Neural Network?

Answer: For tabular telemetry data, ensemble tree models like Random Forest systematically outperform deep neural networks. Random Forest is invariant to feature scaling, robust against noisy synthetic data, provides interpretable Gini feature importances, and has sub-millisecond inference latency ($O(\text{depth} \times N_{\text{trees}})$), which fits within tight blockchain consensus slot deadlines.

Q10: Why use the multiplicative formula $\text{Stake} \times \text{Trust}$?

Answer: The multiplicative formulation ensures that both dimensions act as mutual security barriers. If a wealthy node acts maliciously, its low trust ($T \to 0$) neutralizes its selection weight despite massive capital. Conversely, a node with perfect

trust ($T=1$) cannot gain significant consensus influence without staking proportional capital, preventing costless Sybil attacks.

Q11: Why not select validators based on trust alone?

Answer: Selecting purely on trust dismantles the economic Sybil barrier. An attacker could spawn thousands of zero-stake pseudonymous identities, execute trivial benign transactions between them to build high trust, and overwhelm the consensus pool at zero financial cost. Stake is indispensable for Sybil resistance.

Q12: What happened in the 2022 Ronin Network Hack?

Answer: In March 2022, North Korea's Lazarus Group exploited the Ronin Network bridge, draining $\$624\text{ million}$. The bridge required 5 out of 9 validator signatures to authorize withdrawals. Attackers used spear-phishing to compromise 4 Sky Mavis private keys and leveraged an unrevoked November 2021 RPC authorization to obtain the 5th signature from the Axie DAO validator, forging fraudulent cross-chain withdrawals.

3. Mandatory Viva Questions & Answers (Questions 13 to 23)

Q13: Was the Ronin Network a Proof-of-Stake blockchain?

Answer: No. This is a critical technical distinction. The Ronin bridge was a **Proof-of-Authority (PoA) multi-signature consortium** requiring 5 of 9 approved signatures. It did not use open Proof of Stake. The incident illustrates the danger of static authorization and key compromise, not an algorithmic failure of PoS staking.

Q14: What is Proof of Authority (PoA)?

Answer: Proof of Authority is a permissioned consensus mechanism where block creation and validation rights are granted exclusively to designated, pre-approved identities (authorities). Rather than staking economic capital or expending energy, validators stake their legal/organizational reputation. It offers high throughput but introduces centralized trust assumptions.

Q15: How is Role 2's ML model trained and evaluated?

Answer: Role 2 trains a Random Forest Classifier using supervised learning on the 5-feature telemetry matrix (X) logged during simulation. The data is partitioned into train/test splits (e.g. 70/30) or cross-validated. Performance is evaluated using AUC-ROC, Precision, Recall, and F1-score against the hidden ground truth label.

Q16: How do you strictly prevent data leakage between simulation and ML?

Answer: We enforce an architectural data leakage wall. The ground-truth status (`true_label = honest/malicious`) is strictly quarantined as an evaluation-only column. It is stripped from the feature matrix X prior to model training. The ML model observes only runtime telemetry ($uptime, latency, verification, voting, flags$), never the hidden label.

Q17: What is the cold-start problem, and how did your team resolve it?

Answer: The cold-start problem occurs when a new honest validator joins the network with zero historical behavioral telemetry. If assigned a trust score of 0.0, it is starved of ever being selected. If assigned 1.0, attackers can exploit fresh Sybil identities. We resolve this by assigning a **Neutral Bayesian Prior** ($T_{\text{default}} = 0.50$). This gives new nodes a fair initial participation window while requiring them to build positive history.

Q18: What happens if a malicious validator behaves honestly temporarily (on-off attacks)?

Answer: This is known as an on-off or strategic camouflage attack. While temporary honest behavior builds trust, our architecture incorporates asymmetric trust penalties: when an anomalous event or verification failure occurs, trust scores drop precipitously, whereas rebuilding trust requires a prolonged track record of consistent reliability.

Q19: What are the main limitations of your proposed system?

Answer: Key limitations include: (1) The simulation is conducted in a synthetic, simulated environment of 100 nodes and 1,000 transactions, (2) It omits peer-to-peer gossip latency splits and complex network reorg dynamics, (3) It introduces a slight increase in validator selection concentration (Gini coefficient), and (4) Adversaries could attempt to subtly reverse-engineer feature thresholds over extended epochs.

Q20: Is this system ready for immediate mainnet production deployment?

Answer: No. This is a foundational academic research prototype designed to establish empirical proof-of-concept. Deploying on a production L1 mainnet would require formal on-chain cryptographic attestation proofs (e.g. zk-SNARK verifiable ML inferences), hardened protection against adversarial feature poisoning, and comprehensive economic game theory auditing.

Q21: How do you ensure the comparative experiment between Traditional and Trust-Aware PoS is scientifically fair?

Answer: We evaluate both methods against the **exact same underlying validator pool and transaction sequence**. The initial stakes, ground-truth labels, transaction arrival stream, and generated behavioral telemetry noise are identical across both runs. Only the selection formula ($W = S$ vs. $W = S \times T$) varies, isolating it as the sole independent variable.

Q22: What specific research gap does this student project address?

Answer: Existing literature either removes stake entirely (sacrificing Sybil resistance) or relies on brittle, static linear scoring rules. Our project demonstrates how supervised machine learning (Random Forest) can infer an adaptive behavioral trust score that multiplies economic stake ($Stake \times Trust$), providing a controlled empirical evaluation of malicious validator reduction.

Q23: What improvements would you implement in a future iteration?

Answer: Future enhancements include: (1) Implementing Zero-Knowledge Machine Learning (zk-ML) so validators can prove their trust score inference without off-chain trust assumptions, (2) Testing dynamic retraining pipelines under concept drift, and (3) Benchmarking under adversarial collusion cartels.

4. Comprehensive Technical Glossary (30 Core Terms)

Term	Concise Technical Viva Definition
Blockchain	Append-only, decentralized distributed ledger verified via cryptographic hash pointers and P2P consensus.
Distributed Ledger	A synchronized consensus of replicated digital data shared across multiple sites without a central administrator.
Node	A computational device running protocol client software to maintain, broadcast, or validate transactions.
Validator	An active node authorized to propose and attest to blocks by locking collateral or administrative privilege.
Stake	Cryptocurrency collateral locked in an on-chain escrow smart contract to earn validation rights and deter fraud.
Proof of Stake (PoS)	Consensus mechanism where block proposal probability and voting weight are proportional to locked capital.
Proof of Work (PoW)	Consensus mechanism where block proposals require solving non-invertible thermodynamic cryptographic puzzles.
Proof of Authority (PoA)	Consensus mechanism where validation rights are restricted to pre-approved, legally vetted administrative keys.
Consensus	The algorithmic process by which distributed nodes agree on a single deterministic state of transaction history.
Byzantine Fault	An arbitrary or malicious failure mode where a node equivocates, crashes, or broadcasts conflicting data.
Reputation / Trust	A quantified assessment of a node's historical operational fidelity and adherence to network rules.
Trust Score (T_i)	A continuous normalized value $T_i \in [0, 1]$ output by an ML model indicating confidence in a node's integrity.
Validator Selection	The protocol algorithm that determines which node receives the right to propose the next block in a slot.
Slashing	The automated confiscation and burning of a validator's stake upon cryptographic proof of double-signing or fraud.
Uptime	The proportion of operational heartbeats successfully delivered by a node during an evaluation epoch.
Latency	Elapsed time in milliseconds between transaction broadcast and validator signature verification.
Feature	An individual measurable property or telemetry signal used as input for an ML classification model.
Ground Truth	The true, unobserved state ($1=\text{Honest}, 0=\text{Malicious}$) used exclusively for model evaluation.
Classification	A supervised ML task where a model maps input feature vectors into discrete categorical class probabilities.
Random Forest	An ensemble learning method that constructs a multitude of decorrelated decision trees via bootstrap bagging.

Precision	The fraction of predicted positive instances that are truly positive ($\text{TP} / (\text{TP} + \text{FP})$).
Recall	The fraction of actual positive instances that were successfully identified ($\text{TP} / (\text{TP} + \text{FN})$).
F1-Score	The harmonic mean of precision and recall ($2 \cdot \frac{P \cdot R}{P + R}$), balancing false alarms and missed detections.
Cold Start	The state of uncertainty when assigning initial reputation to newly registered nodes with zero history.
Data Leakage	The accidental exposure of target labels or post-hoc information to an ML model during training.
Sybil Attack	An attack where an adversary creates multiple pseudonymous identities to gain disproportionate control.
Collusion	Coordinated covert cooperation between validators to reorder transactions, censor rivals, or inflate trust.
Gini Coefficient	A statistical measure of inequality across a distribution (0 = perfect equality, 1 = absolute monopoly).
MSR	Malicious Selection Rate: the proportion of total consensus blocks proposed by malicious validators.
HSR	Honest Selection Rate: the proportion of total consensus blocks proposed by honest validators ($1 - \text{MSR}$).

5. Academic Integrity Guardrails: "What NOT to Claim"

Critical Academic Boundaries for Viva Defense

Examiners will rigorously penalize overclaiming. Maintain the following intellectual boundaries:

- **DO NOT claim the Ronin Hack was a Proof-of-Stake failure:** Ronin was a 5-of-9 Proof-of-Authority multisig bridge. It exemplifies key compromise, not PoS design failure.
- **DO NOT claim stake guarantees honesty:** Capital skin-in-the-game creates financial alignment but cannot prevent key theft or mercenary attacks.
- **DO NOT claim our selection formula reproduces Ethereum Gasper:** Production Ethereum uses discrete 32 ETH effective balances and RANDAO lottery beacons; we use a continuous lottery abstraction.
- **DO NOT claim trust scores are pure Bayesian probabilities:** Random forest tree averages represent ensemble class fractions unless calibrated via Platt scaling or isotonic regression.
- **DO NOT claim ML guarantees zero malicious selections:** ML is probabilistic. It slashes malicious selection probability from $\approx 20\%$ down to $<5\%$, but cannot achieve theoretical zero.
- **DO NOT claim this system is production-ready:** Our prototype is a controlled simulation of 100 nodes and 1,000 transactions; it omits P2P network splits and gas auction dynamics.